Project-3

Artifacts stored at Git

https://github.com/singh-ajmer-git/devopsbuild.git

Step-1 Clone the repo

https://github.com/sriram-R-krishnam/devops-build

git clone https://github.com/sriram-R-krishnan/devops-build.git

```
root@ip-172-31-30-99:/home/ubuntu/devops-build/build# ls
_redirects  asset-manifest.json  favicon.ico  index.html  logo192.png  logo512.png  manifest.json  robots.txt  static
root@ip-172-31-30-99:/home/ubuntu/devops-build/build#
```

Dockerfile-

```
root@ip-172-31-30-99:/home/ubuntu/devops-build/build# cat Dockerfile
FROM public.ecr.aws/nginx/nginx:alpine
RUN rm -rf /usr/share/nginx/html/*
COPY . /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
root@ip-172-31-30-99:/home/ubuntu/devops-build/build#
```

Docker Compose

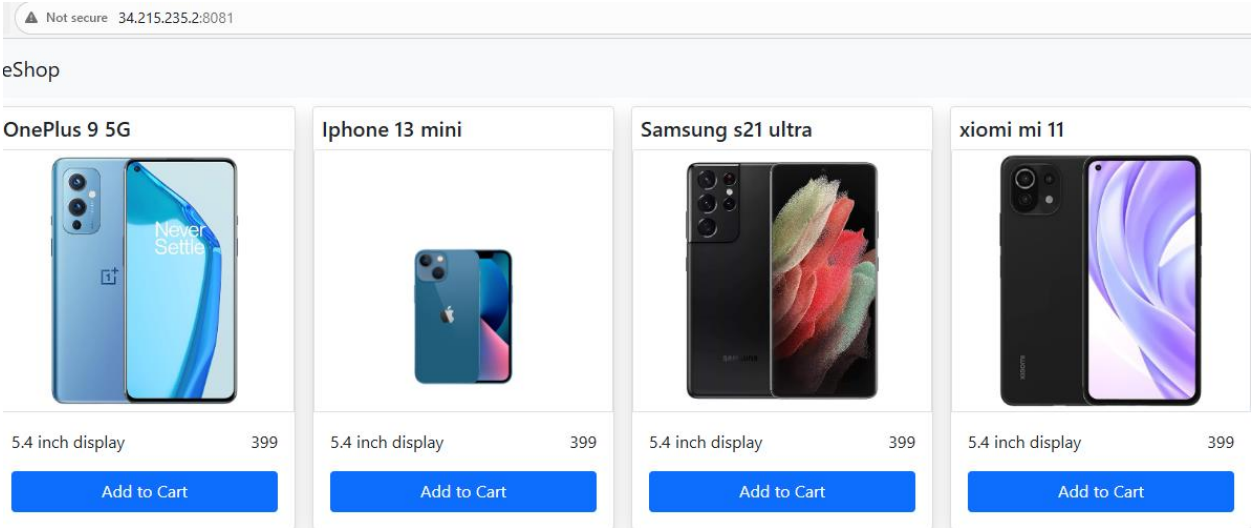
```
root@ip-172-31-30-99:/home/ubuntu/devops-build/build# cat docker-compose.yml
version: "3.8"

services:
  frontend:
    build: .
    container_name: testdev_container
    ports:
      - "8081:80"
    restart: unless-stopped
root@ip-172-31-30-99:/home/ubuntu/devops-build/build#
```

Execute the command to build the image and run the container.

docker compose up -d –build

```
root@ip-172-31-30-99:/home/ubuntu/devops-build/build# docker images

IMAGE                                                   ID             DISK USAGE   CONTENT SIZE   EXTRA
524781515606.dkr.ecr.us-west-2.amazonaws.com/trend-repo:v3   751796a71ea5        111MB          35MB
build-frontend:latest                                   3373527c33f9         96MB         26.7MB    U
hello-world:latest                                      85404b3c5395       25.9kB         9.52kB
jenkins/jenkins:lts                                     23b84d1066db        802MB         288MB
trend:v1                                                f055c9fa5220        111MB          35MB
trend:v3                                                751796a71ea5        111MB          35MB
root@ip-172-31-30-99:/home/ubuntu/devops-build/build# docker ps
CONTAINER ID   IMAGE            COMMAND                  CREATED         STATUS         PORTS                                       NAMES
9178f13cbe04   build-frontend   "/docker-entrypoint.…"   6 minutes ago   Up 6 minutes   0.0.0.0:8081->80/tcp, [::]:8081->80/tcp   testdev_container
root@ip-172-31-30-99:/home/ubuntu/devops-build/build#
```

Expose the container and access it from public to test it locally.



Github repo - https://github.com/singh-ajmer-git/devopsbuild.git

Have added files as .gitignore & .dockerignore to ignore files that are not required to push.

git init

git remote –v

git remote origin add https://github.com/singh-ajmer-git/devopsbuild.git

git add .

git commit –m "update code to push on git hub"

git branch

git checkout –v dev

git push origin dev

DockerHub

Dev -- singajmihub/dev_devopsbuild -Public repo

Prod -- singajmihub/prod_devopsbuild -Private repo

Name	Last Pushed	Contains	Visibility	Scout
singajmihub/react-app-prod	about 3 hours ago	IMAGE	Private	Inactive
singajmihub/react-app-dev	about 4 hours ago	IMAGE	Public	Inactive

Have created the 3 directories to categories

```
root@ip-172-31-30-99:/home/ubuntu/devops-build# ls
build  infra-code  monitoring
```

Build for application source code and build, deploy and jenkinsfile.

Infra-code for terraform code to build the ec2 infra that has docker installed on it.

Monitoring the Prometheus and alert manager to monitor the app.

```
root@ip-172-31-30-99:/home/ubuntu/devops-build/build# ls -a
.   .dockerignore  .gitignore  Jenkinsfile  asset-manifest.json  deploy.sh           favicon.ico  logo192.png  manifest.json  static
..  .git           Dockerfile  _redirects   build.sh             docker-compose.yml  index.html   logo512.png  robots.txt
root@ip-172-31-30-99:/home/ubuntu/devops-build/build#
```

Build – here we have files & directory that contain the source code and build file such as build.sh, deploy.sh, .dockerignore , .gitignore, Jenkinsfile. We set instructions for these files in Jenkinsfile pipeline.

Monitoring - have files to setup and capture the status of app, for monitoring setup.

```
root@ip-172-31-30-99:/home/ubuntu/devops-build/monitoring# ls
alert.rules.yml  alertmanager.yml  prometheus.yml
```

Infra-code – have files for ec2 set-up as required for project as have provisions for this infra via terraform.

The infra with rules that SSH can go only from my IP and application can be accessible via port 80 to all.

```
root@ip-172-31-30-99:/home/ubuntu/devops-build/infra-code# ls
main.tf  provider.tf  terraform.tfvars  variables.tf
```

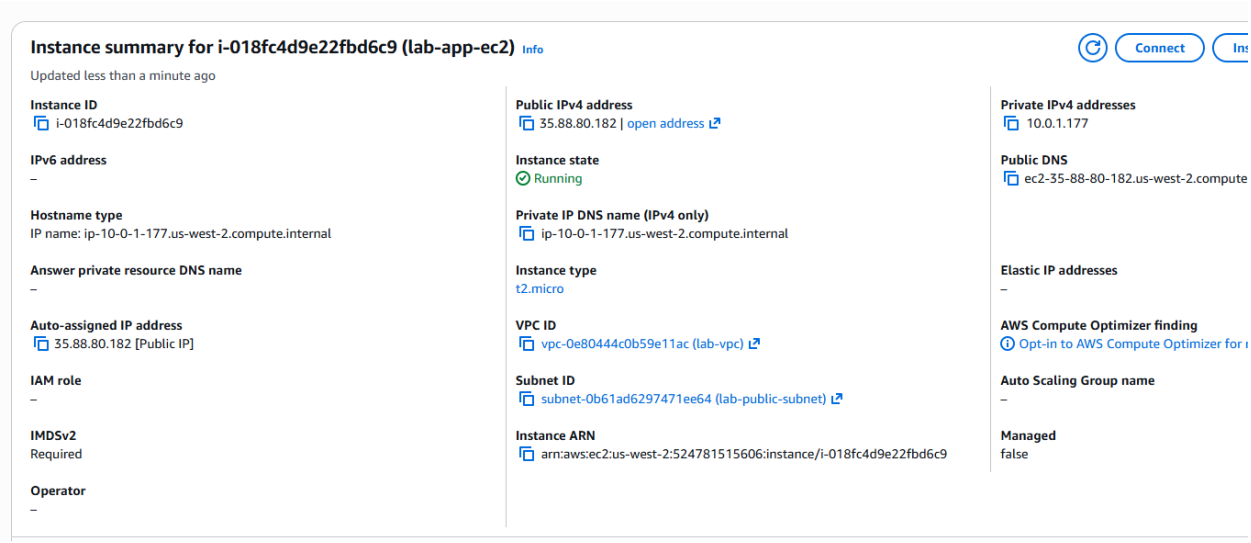
Terraform init

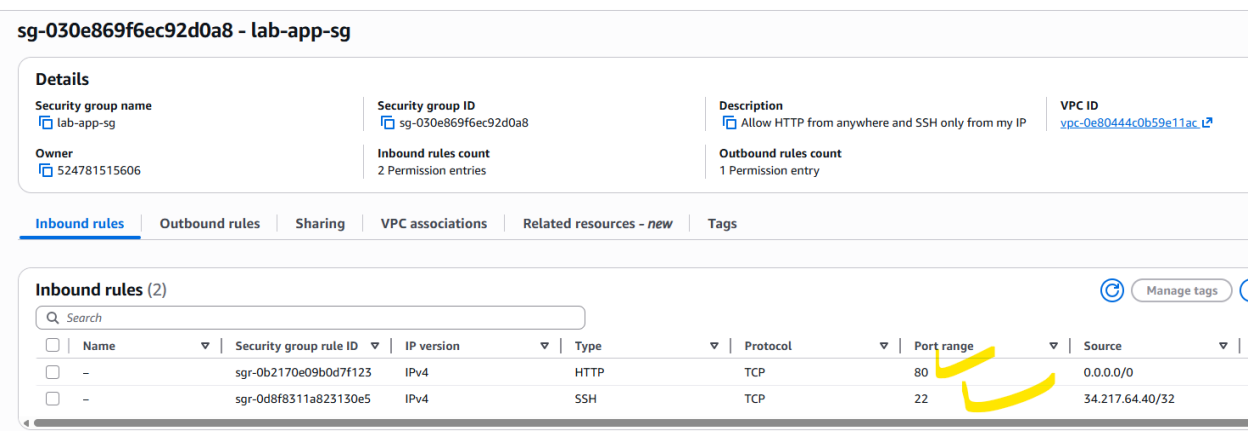
Terraform fmt

Terraform validate

Terraform plan

Terraform apply –auto-approve





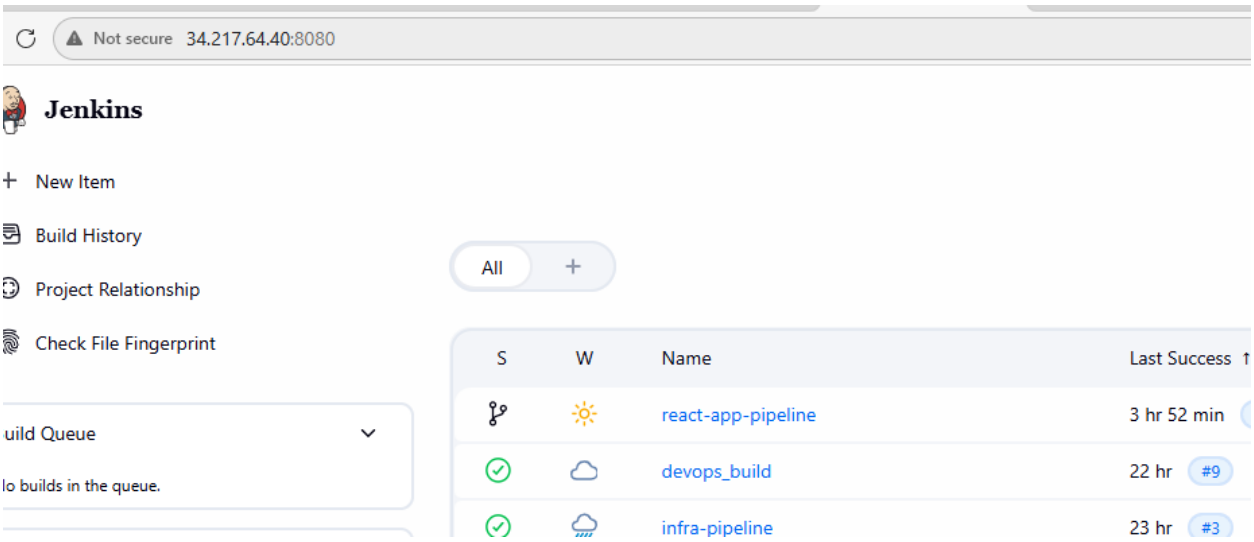
Validate that can do SSH only from the provided IP in inbound rules

```
root@ip-172-31-30-99:/home/ubuntu/seckey# ssh -i "ubuntu-ssh.pem" ubuntu@ec2-35-88-80-182.us-west-2.compute.amazonaws.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Mar 20 00:40:40 UTC 2026

  System load:  0.0               Processes:             115
  Usage of /:   38.7% of 6.71GB   Users logged in:       0
  Memory usage: 30%               IPv4 address for enX0: 10.0.1.177
  Swap usage:   0%

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

1 update can be applied immediately.
To see these additional updates run: apt list --upgradable

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm


*** System restart required ***
Last login: Thu Mar 19 23:27:48 2026 from 34.217.64.40
ubuntu@ip-10-0-1-177:~$
```

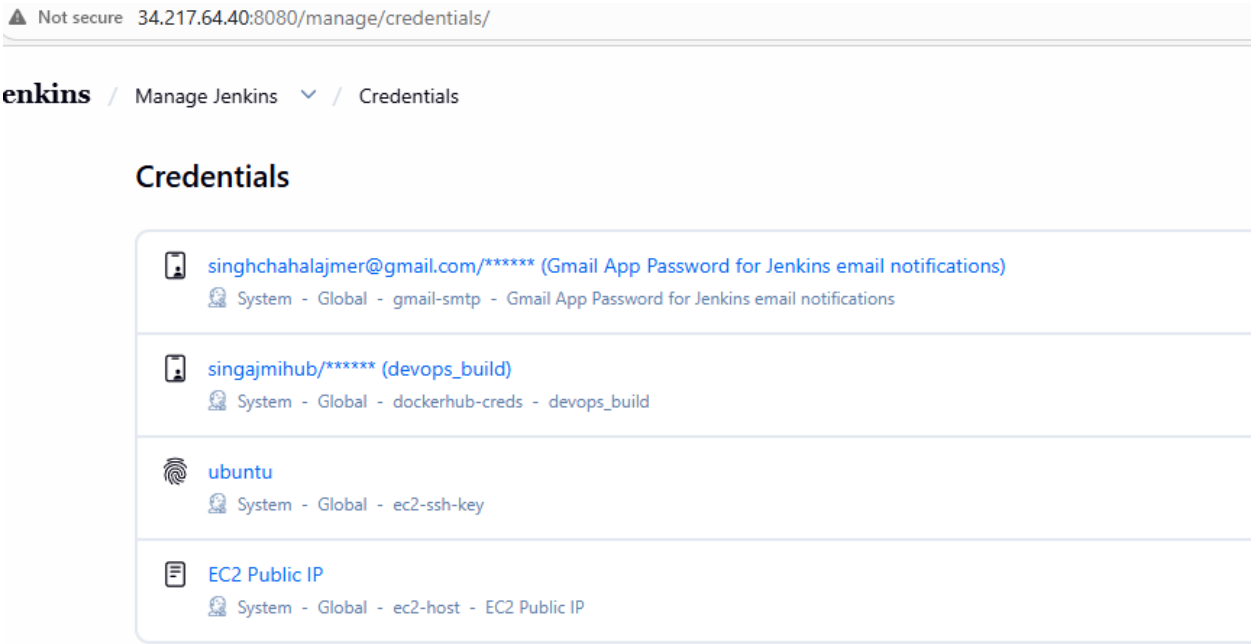
Jenkins - have set-up Jenkins on EC2 machine

```
root@ip-172-31-30-99:/home/ubuntu/devops-build# which jenkins
/usr/bin/jenkins
root@ip-172-31-30-99:/home/ubuntu/devops-build# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
    Drop-In: /etc/systemd/system/jenkins.service.d
             └─override.conf
     Active: active (running) since Thu 2026-03-19 21:10:20 UTC; 3h 8min ago
   Main PID: 7143 (java)
      Tasks: 48 (limit: 9301)
     Memory: 899.8M (peak: 909.9M)
        CPU: 1min 23.999s
     CGroup: /system.slice/jenkins.service
             └─7143 /usr/bin/java -Xms1g -Xmx4g -XX:+UseG1GC -Djava.awt.headless=true -jar /usr/share/java/jen

Mar 19 21:10:17 ip-172-31-30-99 java[7143]: 2026-03-19 21:10:17.480+0000 [id=39]        INFO        jenkins.In
Mar 19 21:10:17 ip-172-31-30-99 java[7143]: 2026-03-19 21:10:17.558+0000 [id=36]        INFO        jenkins.In
Mar 19 21:10:18 ip-172-31-30-99 java[7143]: 2026-03-19 21:10:18.195+0000 [id=37]        INFO        h.p.b.g.Gl
Mar 19 21:10:20 ip-172-31-30-99 java[7143]: 2026-03-19 21:10:20.193+0000 [id=37]        INFO        jenkins.In
Mar 19 21:10:20 ip-172-31-30-99 java[7143]: 2026-03-19 21:10:20.195+0000 [id=37]        INFO        jenkins.In
Mar 19 21:10:20 ip-172-31-30-99 java[7143]: 2026-03-19 21:10:20.566+0000 [id=38]        INFO        jenkins.In
Mar 19 21:10:20 ip-172-31-30-99 java[7143]: 2026-03-19 21:10:20.605+0000 [id=37]        INFO        jenkins.In
Mar 19 21:10:20 ip-172-31-30-99 java[7143]: 2026-03-19 21:10:20.696+0000 [id=37]        INFO        jenkins.In
Mar 19 21:10:20 ip-172-31-30-99 java[7143]: 2026-03-19 21:10:20.775+0000 [id=30]        INFO        hudson.lif
Mar 19 21:10:20 ip-172-31-30-99 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
```

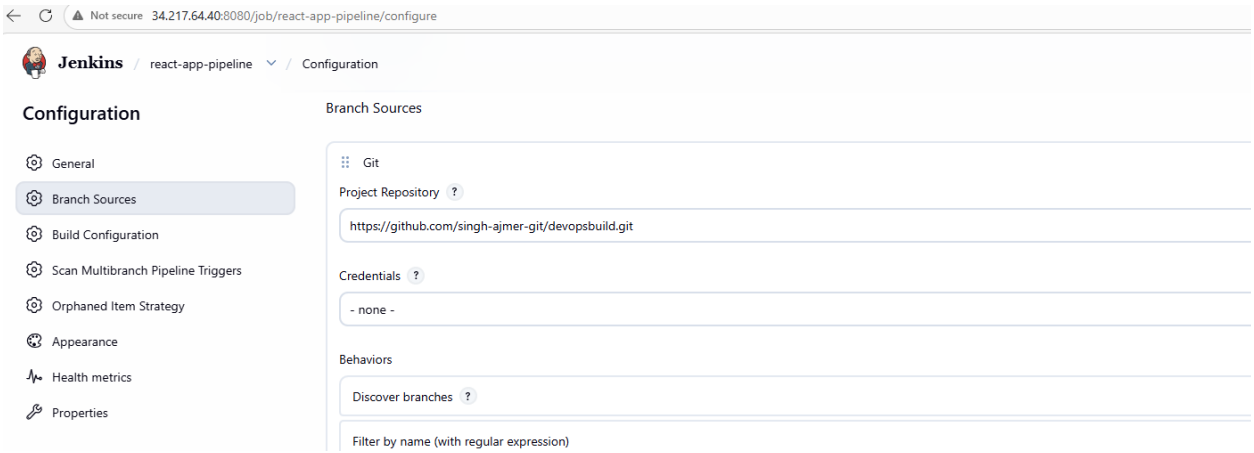
Accessible via web -



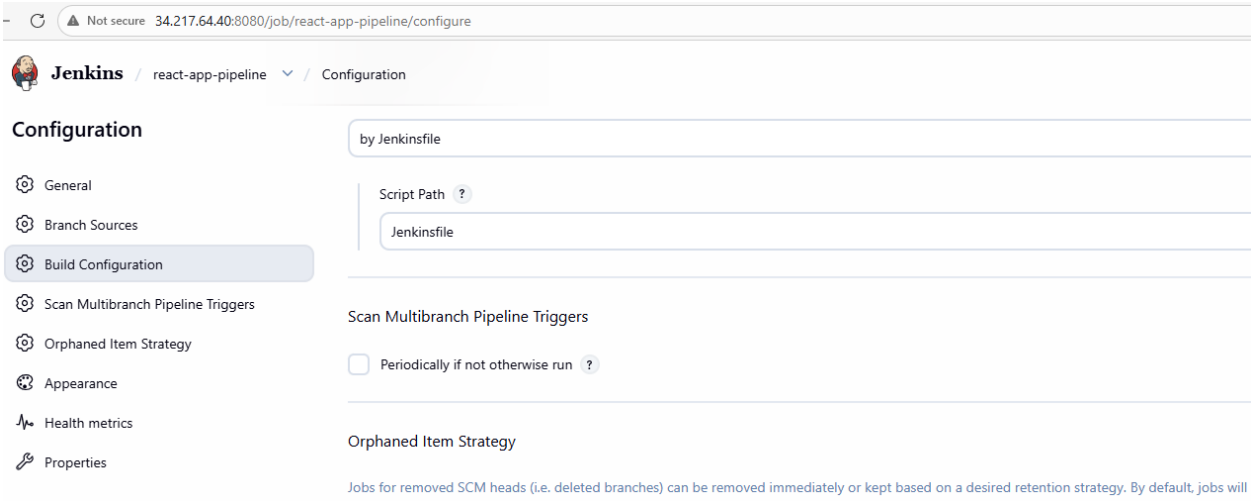
Did setup of Jenkins plugins and credentials for the project requirements.



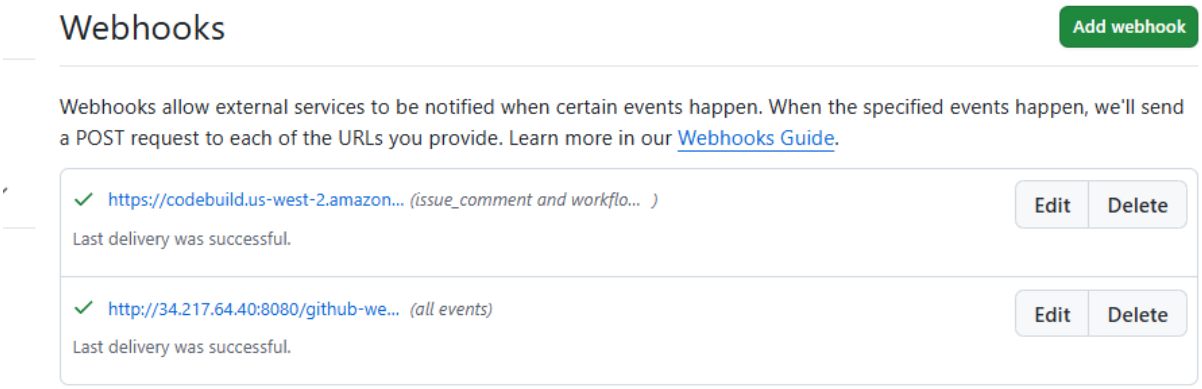
Defies Multibranch Pipeline – as source is Git repo, https://github.com/singh-ajmer-git/devopsbuild.git,



Jenkinsfile script to be execute-



Set-up the Webhook



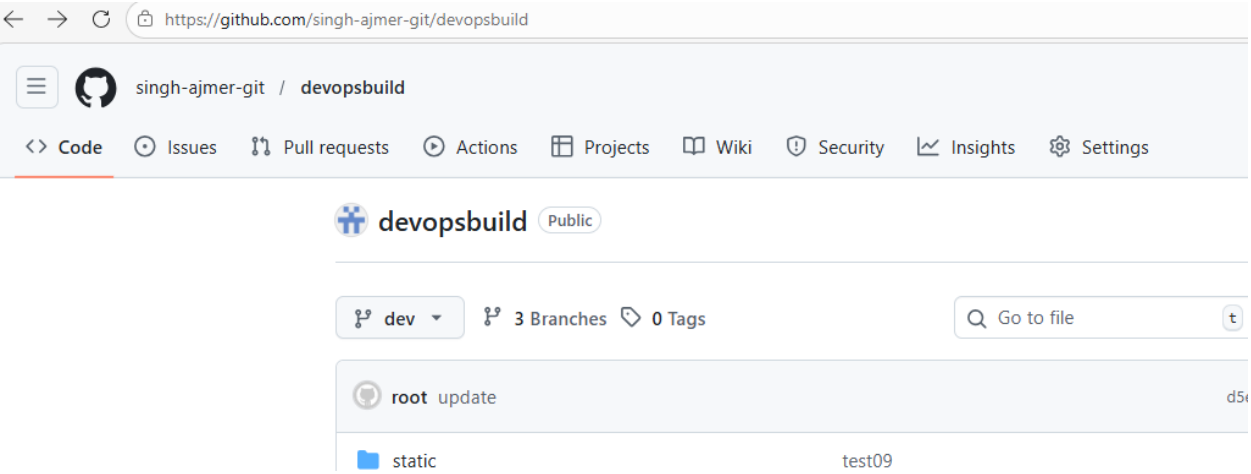
The code pushed from local to git dev branch as followed the commands

git add .

git commit –m “ code update to dev branch”

git checkout dev

git push origin dev



Post that the pipeline trigger and build the image and pushed it to docker hub singajmihub/react-app-dev (Public Repo)

[singajmihub/react-app-dev general]

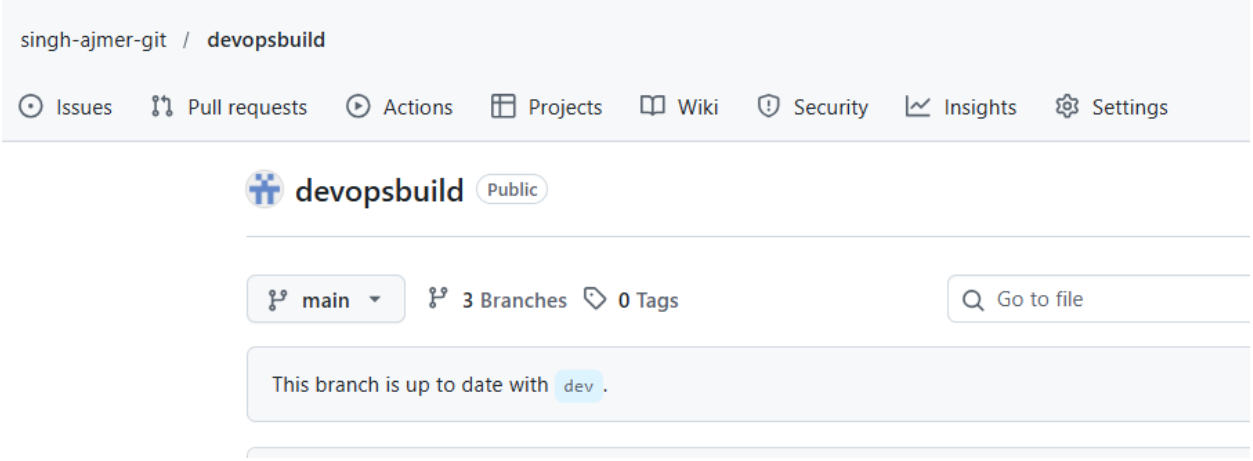
Merged the dev branch to main

git checkout main

git pull origin main

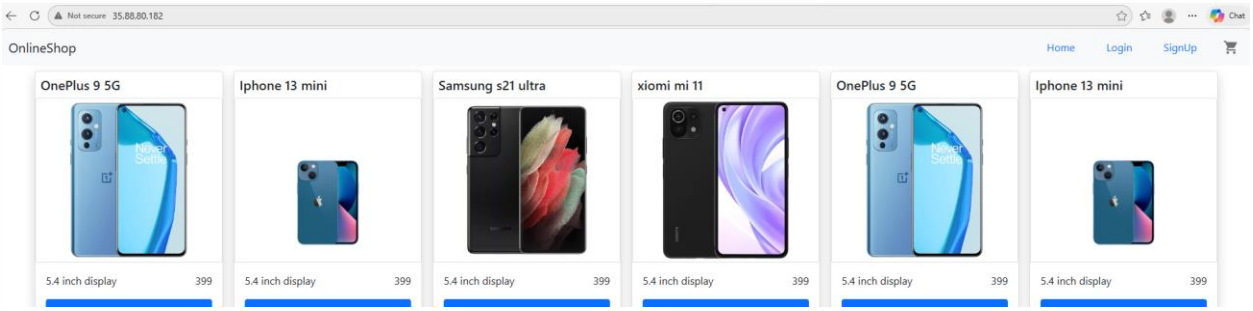
git merge dev

git push origin main



In Jenkins - **react-app-pipeline** pipeline trigged and build the image and pushed to singajmihub/react-app-prod general (Private repo)

In continues of build, this pipeline has deployed the image on EC2, as did ssh to ec2 and run app container, application running fine.



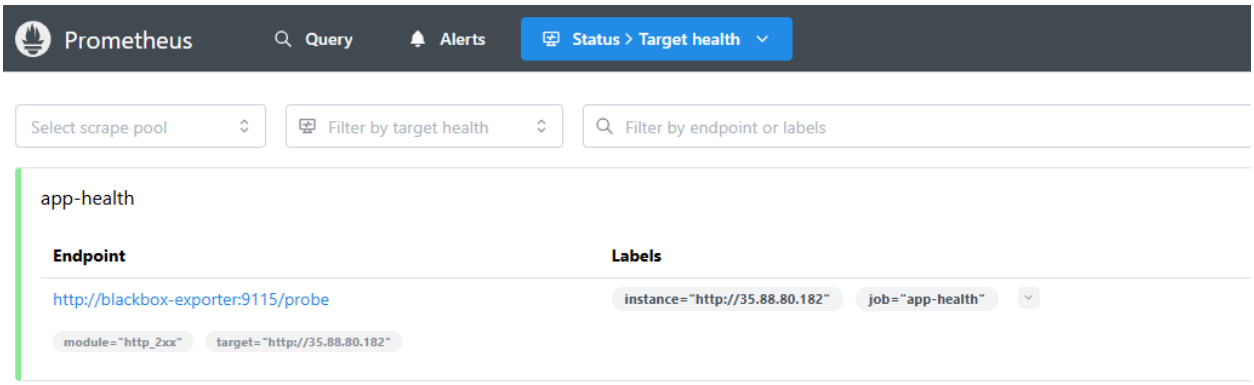
Monitoring -

```
root@ip-172-31-30-99:/home/ubuntu/devops-build# docker ps
CONTAINER ID   IMAGE                    COMMAND                  CREATED          STATUS          PORTS                                                   NAMES
65dee12f0fbd   prom/alertmanager        "/bin/alertmanager -…"   54 minutes ago   Up 54 minutes   0.0.0.0:9093->9093/tcp, [::]:9093->9093/tcp             alertmanager
aa9fceb5acbc   prom/prometheus          "/bin/prometheus --c…"   2 hours ago      Up 50 minutes   0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp             prometheus
49eaa4e4979c   prom/blackbox-exporter   "/bin/blackbox_expor…"   2 hours ago      Up 50 minutes   0.0.0.0:9115->9115/tcp, [::]:9115->9115/tcp             blackbox-exporter
root@ip-172-31-30-99:/home/ubuntu/devops-build#
```

Have setup prometheus along with blackbox-exporter & alertmanager

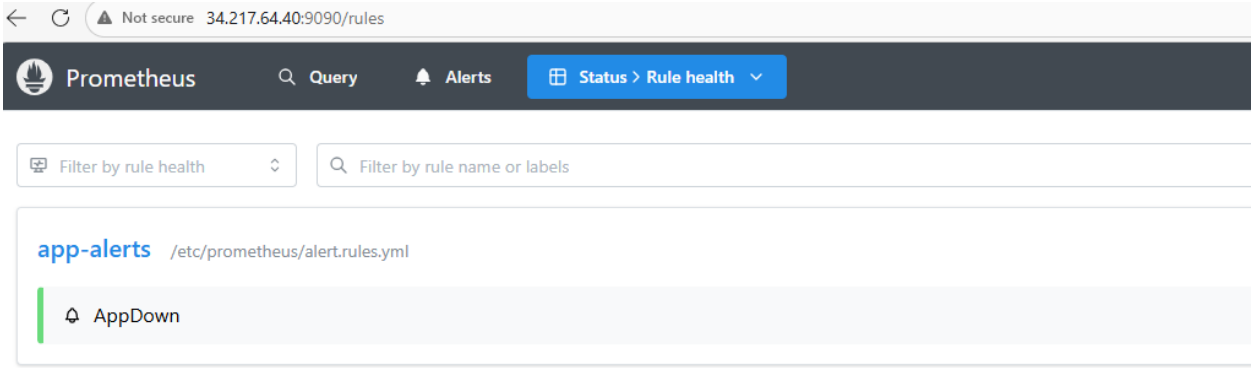
Blackbox Exporter → Prometheus → Alertmanager → Email/notification

Target: your React app URL



Email alert – Stop the app

```
ubuntu@ip-10-0-1-177:~$ sudo su
root@ip-10-0-1-177:/home/ubuntu# docker ps
CONTAINER ID   IMAGE                                COMMAND
ade4efa18c98   singajmihub/react-app-prod:latest    "/docker-entrypoint.…"
root@ip-10-0-1-177:/home/ubuntu# docker stop react-app
react-app
root@ip-10-0-1-177:/home/ubuntu#
```



Wait for an minute

[FIRING:1] (AppDown http://35.88.80.182 app-health critical) Inbox ×

AJ **singhchahalajmer@gmail.com**
to me

1 alert for

View In Alertmanager

[1] Firing

Labels
alertname = AppDown
instance = http://35.88.80.182
job = app-health
severity = critical
Annotations
description = The application is not reachable
summary = Application is DOWN
Source